Troll Farm — The D-Series Atlas

A reader's guide to every numbered experiment of the Legend top-3 cycle

Snapshot 2026-07-27 · ledger vol 1+2 · `docs/CONSTRAINTS.md` · `docs/STATE.md`

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Snapshot date: 2026-07-27. A reader's guide to every numbered experiment of the Legend top-3 research cycle. Chronology lives in the ledger volumes (`data/analysis/live-agent-6553250/legend-top3-expe`; topic-sorted conclusions in `docs/CONSTRAINTS.md`; live state in `docs/STATE.md`. This atlas is a navigation snapshot, not a living record.

1 What a D-experiment is

Each `Dnnn` is one frozen-protocol experiment in the project's laboratory notebook. A letter suffix (`D133b`, `D41c`) is a protocol revision or repair of the same experiment. Every experiment leaves the same artifacts in `data/analysis/live-agent-6553250/`: a **protocol** (question, method, and pass/fail thresholds written before running), a **lock** (checksums of protocol, code, and inputs), the run products (bulk rows on the external `artifacts/` root), and a **result** document ending in an explicit verdict. The discipline that holds it together: thresholds are never adjusted after seeing data, consumed seed ranges are never reused for selection, and negative results are retained as binding constraints. Numbering makes conclusions citable ("remembered-crop returns are closed — D165") and prevents re-running dead ends.

The goal (set 2026-07-18): rank 3 or better in the Legend practice ladder, on a mature read plus confirmation. As of 2026-07-27 the resident (agent 6561795, exact slim Yamo/Orchard, two workers) reads 43/110 at 21.97 against a rank-3 bar of 28.22. The measured disease: the resident leads at turn 100 in most eventual losses and is out-compounded late by opponents who scale to three-plus workers while keeping production alive; top agents reap 24.16% of the crops they create, the resident 0.94% (D101).

2 Prelude — Phases 1–21 (2026-07-16 → 07-18)

Before the D-series, a phase-numbered program recovered the live bot and made the first candidate attempts. Highlights:

Phase	Question	Verdict
Recovery	What is actually live?	Exact source recovered (rank 6/104 @ 26.31), checksummed, made submit default
1–5	Do micro-edits help?	Idle-harvest KEEP; focus bonus KEEP; training constants closed; renewable mothers closed
Terminal fix	Are local verdicts valid?	Referee stall/mercy semantics adopted; one verdict flipped (pre-seed neutral → +0.259)
Stack	Pre-seed + orchard geometry	Arena **+3.0 — PROMOTED** 2026-07-17; slim A/A passed; still the only arena win
3–5	First-turn rollout controller	Local +2.7 → arena reject (21.7 vs 24.1 control)
6–8	29-option first-move library	No opponent-robust activation; turn-one Monte Carlo closed
9	Imitate Escdemon	Passes observationally; collapses at autoregressive integration (52% MOVE)
10	Norxondor's workforce ladder	Reproduces all 8,738 triggers; observable-signature switch fails prospectively (−6.2)
11	Online shared-state Monte Carlo	Teacher +26.1 but 209 ms vs 50 ms budget — undeployable
12–15	Distill trajectory value	Fails held-opponent-family precision every time; native imitation −172.7
16	Resident MOVE residual	Prospective +0.508 at 93 ms p95 — closed
21	Dual-value opponent-crop scoring	All local gates pass → arena **−7.77 REJECT**; b100_e6 evolved candidate +0.12, closed

Lesson carried forward: local gates were not predicting arena outcomes; the cycle moved to architecture research with much harder transfer discipline.

3 Arc A — Curriculum and the deployment bridge (≈D1–D28, 07-19 → 07-20)

Can a neural policy learn this game at all, and can it ship inside CodinGame's limits? A training curriculum was built in five levels; numbering here is partly internal to the rolling narrative (headered experiments begin at D29).

Item	Question	Verdict
PPO from scratch	Learn the debug task raw	12.4% — closed at readiness
BC + teacher-auxiliary	Bootstrap from a scripted teacher	Pure PPO argmax-collapses (18.2%); a 0.10 teacher anchor rescues it (99.3%)
Level 1	Execute one worker's resource plan	Accepted, independently replicated (99.65%)
Level 2	All 8 worker recipes	Accepted (98.6%), discovery + confirmation
Level 3	Two-role renewable loop	Accepted (99.45%); illegal-label mask repaired
Level 4	Recipes × roles jointly	Accepted (99.70%)
L5 D1–D5	Isolated opponent mechanisms	Forager, planter, reaper, funded 2-worker: all accepted
L5 D6–D8	Paid third worker, horizons	Teacher-censoring artifacts; closed at controls
L5 D9	Crop before scale	Accepted — "secure renewable supply before scaling"
L5 D10–D11	Repeated pressure; seed reacquisition	D11 = first task to beat the actor (79.4%); clone+PPO accepted at 97.65% prospective
K1–K2	int8 deployment	K2 accepted: 7.04 ms median inference
V1–V5	Live-protocol integration	Four observation-parity defects fixed; V5 accepted (68,988 B, 17.6 ms warm p95)
YT bench-mark	Train remotely?	9.8× throughput but CUDA/CPU parity fail → local training chosen

Surviving assets: the environment/actor infrastructure, the deployment pattern (int8 + persistent buffers), the teacher-auxiliary recipe — and a deployable curriculum actor whose *field* value was still unproven.

4 Arc B — The domain-shift reckoning (D29–D34, 07-20)

ID	Question	Verdict
D29a/b	Turn-75 spatial option-critic (farm vs resident)	+36.5 on generated maps; int8 candidate built
D29c	Does it activate on real arena states?	8.75% vs 17–67% corridor — closed unsubmitted
D30	Why the −78 shift?	Generator artifact: 6 water cells vs official 12–104; all roots out of support

ID	Question	Verdict
D31	Replay recorded commands as continuations?	Invalid — only 86.85% command match
D32a	TestSession forced-farm A/B	Mean −29; permanent farming and threshold repair closed
D33a/b	Port the referee's real map generator	Accepted — SHA1PRNG + plant-order subtleties; 120/120 exact
D34	Census: 9 complete architectures on official maps	None passes production+suppression; worker count alone ≠ strength

This arc invalidated the old evaluation world and replaced it with the authoritative official-map substrate. Everything after D33 runs on real map geometry.

5 Arc C — Job grammars and bounded overlays (D35–D36, 07-20 → 07-21)

ID	Question	Verdict
D35a	Decode rich players' persistent jobs	Vocabulary strong (94% RENEW+FELL); flat categorical head fails
D35b	One-shot factorized two-worker bundles	Oracle +34.99 but suppression far short
D35c	Add crop-creator provenance to targets	Causal +7.8 — provenance retained as a factor; one-shot closed
D35d	Repeat allocation at job boundaries	+23.1 beyond one-shot; opponent excess still +76 vs +65 ceiling
D36	Same overlays anchored on the resident	+19.6 own / +10.6 margin vs +68/+25 floors — closed

Conclusion recorded: no wrapper composition reaches the field gap; one coherent policy must own opening, supply, funding, production, and suppression end to end.

6 Arc D — Complete-macro teachers (D37–D40, 07-21)

ID	Question	Verdict
D37	Complete factorized macro environment	Environment clean and deterministic; rate initializer fails; transactional semantics invalidation found and fixed
D38	Value jobs by exact TRAIN-bill reduction	Rejected (loses to random); spawn-shack blocker isolated
D39	Add shack evacuation	Narrow reject (+38.2 vs +50); insight: 1→3 promotions gain +295, 1→2 gain −1.3
D40	Deficit-first, else work-conserving rates	**PASSES** +94.6 over D39, all families — frozen as THE teacher

7 Arc E — Learning on D40 (D41–D62, 07-21)

The graveyard of straightforward learning ideas, and two validated interfaces.

ID	Question	Verdict
D41a	Behavior-clone D40	85% vs 99% floor; a parameter-free decoder gets 100% — MLPs can't do lexicographic filters
D41b	Exact-prior + zero residual	Byte-exact reproduction — validated interface
D41c	Residual PPO (temp 4)	Max residual +0.33 vs +4 prior gap → 0/85,128 changed decisions
D41d–f	One-deviation labels, branch selectors, thresholds	46–58% positive vs 55–60% floors — all fail
D41g/h	Linear / tiny-ReLU value filters	Fail the 65%/27% precision pair
D42	194-feature context selector	Fails; one-deviation supervision closed
D43/D44	Sparse binary closed-loop PPO	Learns a uniform bias (probe SD 0.0012); its scores uncorrelated with value
D45/D48	Rate-formula surfaces	Saturated/discontinuous — not locally searchable; CEM blocked
D46/D47	Hard chopper / producer roles	Chopper already implicit in D40; producer role −12.0
D49	Chopper-first reservation order	12 deposit-prediction violations — transactional revalidation required
D50	Population of phase-recombined opponents	Support improves; rich-cohort gates still missed
D51	Workforce-relative whole-controller switching	Only 27.5% ever reach the trigger — funding is the upstream problem
D52a/b	Procedural job market; why TRAIN fails	All 5,142 failures = a PICK spends the bill after the decision
D53a/b	Atomic bill reservation	Fixes worker-2; 168 residuals = shared-surplus oversubscription
D54	Shared per-turn PICK ledger	Transactions fully clean; reach still 25.3% — problem moves to acquisition
D55	Terminal stock-flow diagnostic	91% of blocked bills source-unresolved; LEMON dominates
D56/D57	LEMON source; exact deficit-vector sources	Sources get built; worker-3 reach *falls* (23.8%, 21.3%)
D58	Labor/progress diagnostic	Source planting regresses bills; mining converts — conversion layer is causal
D59	Materialization-only lease	Better per-turn progress, 56% idle labor — entire hand-designed source branch closed
D60	Fixed invest/materialize/liquidate per phase	Oracle +47.4 but violates the crop floor — fixed modes rejected
D61	Options at natural job-batch boundaries	Oracle +57.6, crop-safe, sequencing carries value — **validated interface**
D62	Feed-forward batch-option PPO	Argmax never moves (0/512) — recipe closed; passive field refresh (D61p) readied

8 Arc F — Field evidence, round two (D63–D71, 07-21)

ID	Question	Verdict
D63a/b	What predicts who scales?	Turn-100 economy state (0.97 AUC), not opening geometry (0.48); prediction ≠ value
D64a/i	Gate late scale by the field model	Run invalid via a worker-two tail; oracle only +1.9; diagnosis: missing bill species
D65–D67	Rescue via deposited-seed sources	Planted sources die before payback; lease fails; zero viable cells — closed
D68	Bill-level source portfolio	2/4 recovery; reusable hostile pressure taxes each generation — late rescue closed
D69/D70	Scaler lifecycle archaeology	Universal order establish→receipt→reinvest→worker-3; no minimal opening transaction reproduces it
D71	Closed-loop opening-portfolio environment	Mechanics pass — infrastructure only

9 Arc G — Recurrent search era (D72–D78, 07-21)

ID	Question	Verdict
D72	Recurrent opening portfolios (8 actions)	Oracle +64.1 headroom; explicit seed-action breadth misses by one task
D73	Four-mode recurrent PPO	−1.76 prospective; sacrifices own score symmetrically
D74/D75	One- and two-batch option sequences	Too coarse: 38.5% strict; 16 sequences span 3.5 points
D76/D77	CEM; full-parameter lineage search	Both converge to the balanced plateau (≤1% non-balanced choices)
D78a/b	Is opponent attack imminence observable?	Yes — 0.9307 AUC from spatial features; history adds nothing

Recorded pattern: with four coarse modes and a snapshot state, every optimizer retreats to ”do what the resident does.” The action space, not the optimizer, was the constraint.

10 Arc H — Spatial and target interventions (D79–D85, 07-21)

ID	Question	Verdict
D79	Unconstrained all-Rate spatial scorer	Global trajectory replacement (every policy changes every hash) — closed
D80/D81	One-shot contested-crop promotions	Ubiquity/support failures; lesson: measure authority by decisions, not hashes

ID	Question	Verdict
D82	Semantic responses to threatened crops	Oracle +11.2 but value is state-specific — no deployable fixed arm
D83	Snapshot value model over D82	Captures 10.6% of oracle — closed
D84	Truncated counterfactual Monte Carlo	Best +3.2 vs +5.6 needed, at 210 ms — online MC closed
D85	One-turn defensive salvage on real replays	The resident already makes every lethal joint chop; salvage +0.05 — closed

11 Arc I — The yaichi seed-factory (D86–D95, 07-21)

ID	Question	Verdict
D86	Is renewable mode opening-selectable?	No — 0.58 validation accuracy
D87	Commit to regeneration after fresh harvests	−51.2 active margin; new plants become wood, not an orchard
D88a–c	Decode yaichi's task states	Bank-bootstrap seed-factory architecture confirmed 10/10
D89	Reproduce the factory on the resident	Production spectacular (+79.4) but opponent +82.9 — safety reject
D90	Lineage boundaries; ATTACK state	Ablation inert; ATTACK is an endgame bank blockade
D91	Select factory activation by map	Development +31 but 5/16 maps — transfer fail
D92	Factory + dual-value denial	Trained-role denial arrives too late (−5.6)
D93	Can the factory fund worker 3?	Zero legal TRAIN turns in 256 tasks — it's a wood economy, not a funding economy
D94	Late existing-stock funding bridge	Trains in 147 tasks and loses 91.6 — late grafts closed
D95	Rank-one scaler archaeology	Coordinated funding is shared; no universal hand-written grammar — learned-representation requirements recorded

12 Arc J — Joint assignments and the q6 program (D96–D147, 07-21 → 07-23)

The largest arc: a validated coordination interface, a compact action bank, and roughly thirty supervised selectors that all failed the same way.

ID	Question	Verdict
D96	Worker-factorized four modes	+0.8 incremental vs +5 — worker context isn't the missing piece
D97	Concrete collision-safe JOINT assignments	Oracle +36.9; joint adds +9.2 beyond best-single — **validated interface**

ID	Question	Verdict
D98–D100	Pair scorers, replacements, residuals	Near-misses and activity failures; static selection over the bank is dead (rank correlations 0.10–0.22)
D101	Production/suppression role archaeology	□ The gap: top-3 reap 24.16% of created crops, resident 0.94%; suppression already competitive
D102/D103	Transfer D40 wholesale; phase decomposition	−48.4; opponent excess accrues 14/39/47% across phases — duration matters
D104–D106	Expert-proposal union; q4/q6 quantization	Union keeps 86–88% of the joint oracle; q6 bank frozen at 9,180 bytes with fresh-map replication (+32.0)
D107	Bounded online controller preflight	Four-use oracle +35.2, crops/workforce exact — interface ready
D108/D109	Recurrent masked q6 PPO; 4× duration	−0.74 then −0.15; family patterns rotate across panels — the family-robust objective question is posed (and skipped)
D110/D111	Random linear one-use policies; lineage search	Abstention calibration is the frontier; fitness doesn't transfer
D112/D113	Dense exact-continuation teacher	Coverage semantics fixed; teacher +36.8, all families positive
D114–D118	Ridge, MLP, WAIT-softmax, factorized gates, soft targets	All fail; ranking (not timing) is the bottleneck
D119–D127	Long fits, info floors, calibration, tail attribution	First fit+validation pass appears; held support protocol fails; 82/98 losses are ranking errors with +891 recoverable
D128/D129	Absolute anchors; safety heads	Ranking and safety must not share a scalar; compositions 0/60
D130/D131	Pairwise loss; all-seed audit	Fit statistics *anti-predict* transfer (r = +0.89 wrong way)
D132/D133/DY39	YT distributed corpora	Byte-exact parity; eight independent 16-map blocks built
D134–D138	Out-of-fit block selection; gates; calibration	Selection design validated; every learner still fails the veto
D140–D143	8-block selection, balanced losses, dual gates	Best ever: +3.06 at 39.65% strict vs 40% floor — **one-use gate tuning closed**
D144–D146	Offline two-intervention Monte Carlo	+4.15 incremental; winning pairs genuinely joint (second action +27.3); early-immediate schedules carry it
D147	Live-interface feature replay	Byte-exact — safe to scale

13 Arc K — Exact-value teachers and the substrate verdict (D148–D158, 07-22 → 07-23)

ID	Question	Verdict
D148a/b	Fresh-map priority joint teacher (YT)	Hidden two-use transfer +4.11, robust in all families
D149a/b	Imitate its best pairs	9.5–9.8% rank accuracy — hard-argmax entropy, closed

ID	Question	Verdict
D150	Is near-tie supervision available?	16.5% vs 20% — collect counterfactuals instead
D151/D152	Exact conditional-second values (YT)	+36.8 on active states — the strongest teacher signal of the program
D153–D156	Fitted value, abstention, slices, memory, lookup	Everything fails map-fold transfer (+2 held vs +14–17 train); confidence anti-correlates
D157	Frontier audit	The skipped D109 family-robust objective is the one open branch; D158 frozen
D158	Recurrent q6 PPO relaunch	**Stopped by the baseline-dominance audit** — the whole D40/q6 world can't beat the resident

14 Arc L — The resident-native pivot (D159–D168, 07-23 → now)

ID	Question	Verdict
D159	Refresh the loss mechanism on 200 games	Catastrophe tail replicated (11% of games, 58% of negative mass); attack angles ranked
D160	Does the resident ever afford worker 3?	Never — zero affordability windows in 195 games; funding is policy, not luck
D161	Same-panel dominance arithmetic	Full q6 oracle only +3.4 (n.s.) vs resident — substrate closed; resident-anchoring becomes law
D162	Bounded reserve/route/commit options	Can't fund worker 3 (5/128 best) — but the crop-safe option envelope is +12.7 with zero regressions
D163	Do fixed components transport?	No — fruit/iron/protection all nonpositive on a disjoint panel
D164	Current-field macro-transition audit	New motif: producer→suppressor→producer cycling, 72% of top-5 games, resident 11%
D165	Return to the remembered crop?	Zero support in 1,024 tasks — the old crop is always gone
D166	Is the return one command?	No: multi-step acquisition journeys, median 16 turns; single-verb controllers closed
D167	Are the journeys regular?	**Yes: BANK_SEED frozen-eligible** (135/135 local, 71.4% top-5 field); field agents pre-carry seeds through suppression 45%, resident 0%
D168	Does executing the return causally help?	*Running now* — bounded option vs exact resident, preregistered value/tail gates

15 Inflection points

- **D30** — our generated maps were unlike real maps; weeks of results invalidated, honest substrate built (D33).

- **D40** — a teacher economy good enough to learn from.
- **D97** — joint two-worker coordination has real, measured value.
- **D101** — the gap is production *persistence*, not suppression.
- **D161** — the alternative-economy substrate is weaker than the bot we already have; everything pivots resident-native.
- **D164→D167** — the missing field behavior distilled into one frozen-eligible option.

16 Lesson learnt — what rich players' persistent jobs taught us (D35a and descendants)

D35a decoded two frozen partitions of rich-player replays (a 12-game discovery and a 9-game confirmation split) into **persistent worker jobs**: multi-turn units of intent — "this worker is renewing the farm", "this worker is felling and banking wood" — rather than per-turn commands. It is one of the program's foundational results because both the *finding* and the *representation failure* shaped everything after it.

What the decode found. The job vocabulary explains essentially all of what strong players do: direct productive coverage 100% in both partitions, all-unit coverage 98.1%/99.5%, and 96.4%/99.0% of MOVE commands resolvable as travel *for* a job rather than wandering. The median job runs six-to-seven turns — strong play is *committed*, not per-turn reactive. In 62.7%/63.5% of multi-worker turns the workers hold **distinct roles**, and just two job types — RENEW (renewable farm work) and FELL (chop-and-bank) — account for **~94% of all non-idle activity**: the producer/producer/chopper field mechanism in its rawest form.

The representation lesson. The obvious encoding — one flat categorical over whole-team job signatures — failed its own gate: the top 32 team signatures covered only 86.56% of discovery turns against a 90% floor, because teams of one to seven workers generate 176 signatures and a flat head would have to encode *team size* into *class identity*. The verdict distinguished abstraction from representation: **keep the job abstraction, discard the flat joint head, and factorize per worker** — centralized assignment, collision-aware targets, deterministic banking/completion, and a separate global TRAIN decision. That factorized interface became D35b–d (bundles, provenance, repetition), then D97's validated joint assignments, and ultimately the q6 proposal ABI.

What the lesson did *not* license. Hard-coding the observed roles failed both ways: a forced persistent chopper was inert (D46 — the D40 teacher already chooses FELL_BANK whenever legal) and a forced persistent producer lost 12 points (D47) — the roles strong players exhibit are *emergent from scheduling*, not rules you can bolt on. And the later field audits sharpened the picture: what actually separates the top cohort is not holding roles but **cycling** them — the same worker produces, suppresses, then produces again (D164, 72% of top-5 games), with production *persistence* through interruptions as the real differentiator (D101: they reap 24% of what they plant; the resident 0.94%). The D-series' current thread — successor jobs, BANK_SEED returns, D168's bounded option — is the direct descendant of D35a's job abstraction applied to that cycling.

17 Model architectures and situation encodings

Three encoding families and half a dozen model families recur across the arcs. This section describes them concretely.

The raw situation. Every encoding starts from the same game state: a rectangular board (up to 22×11 cells) of terrain (soil/water), trees (species PLUM/LEMON/APPLE/ BANANA/IRON-ore analogues, growth stage, hit points, ripe fruit), units (per-troll move speed, carry capacity, hp, chop power, cargo), both players' banked inventories and scores, and the turn number. The referee reveals all of it every turn — the encodings differ in what they make *learnable*, not in what is visible.

Family 1 — spatial planes (the curriculum actor, Arc A). The Rust environment serializes the situation as a **104-channel × 11 × 22 uint8 tensor**. Per-cell planes encode validity (channel 0 doubles as the board mask), terrain, tree species/stage/fruit, and unit positions; scalar facts are *broadcast* as constant planes — e.g. channels 56–61 own inventory and 62–67 opponent inventory (values quantized to uint8 at a ≈1/30 scale), channels 74–78 the selected unit's move/carry/hp/chop/free-capacity stats — and explicit BFS *distance planes* give the network pathfinding for free (added after pure PPO failed to learn it). Actions are likewise spatial: **13 action planes × 11 × 22** — a masked categorical over (action-type, cell), with the legality mask computed by the referee-exact environment.

The curriculum actor architecture (`SpatialActorCritic`): a 3×3 conv stem 104→16 channels → **four residual blocks** (width 16, two 3×3 convs each) → an actor head (1×1 conv to the 13 action planes, logits flattened and masked) and a critic head (validity-masked global average pool → Linear 16→64 → tanh → Linear→1). Roughly 35k parameters — deliberately tiny, because deployment must fit CodinGame: the accepted pipeline quantizes it to **int8 inside a generated Rust kernel** (max logit drift 0.0000687, 512/512 identical masked choices) and, after the K2/V5 pattern of persistent workspaces and reused buffers, runs a two-worker inference in **7.04 ms median / 17.6 ms warm p95** within a 68,988-byte single-file source. Training used PPO with a 0.10-weight teacher-auxiliary loss (pure PPO deterministically argmax-collapses) and legal-action masking throughout.

Family 2 — scalar situation vectors (selector era). Where a decision attaches to a *moment* rather than a cell, situations are flat feature vectors, always built from observable state only (opponent identity is deliberately excluded — the submitted agent cannot see it): the 56-feature deployable state of D61's option probes, the **64-field state vector** of the q6 program (global economy, workforce, phase, remaining intervention budget, previous-intervention kind), the 139/62/44-feature turn-100 economy models of D63, and D78's spatial-relational add-ons (target condition, worker-to-crop distance, occupancy) that lifted opponent-attack prediction to 0.9307 AUC. The recurring finding: these snapshots *predict behavior* well and *transfer value* poorly — every fitted value model on them collapsed one map-fold away (D153: +14–17 in fit, +1.8 held).

Family 3 — proposal/action vectors (the q6 ABI). Interventions are encoded per *candidate action*: a **379-field proposal vector** (job kind, target class and crop provenance, ETA, predicted deposit and rate, ownership, encoded cell geometry, and the 64-expert endorsement pattern) concatenated with the 64-field state = the 443-feature rows of the D148/D151 corpora. The proposal *generator* is itself a model: the bank of 64 frozen D98 linear scorers, quantized to 6 bits per coefficient (**9,792 coefficients in 9,180 base85 bytes**), whose per-root proposal union retains 86–88% of the exact joint oracle — a compact action basis rather than a policy.

Model families tried on these encodings, smallest to largest: linear probes (224 and 379 weights — D61/D110), a 12-unit echo reservoir with a 52-parameter trained readout (D76), a fully-trained 1,072-parameter recurrent controller (D77), tiny MLPs of 6–7k parameters (D115 ReLU classifier; D153's state64+action379→16→1 value net), the factorized 6,626-parameter proposal-ranker + 689-parameter act/wait gate (D117–D143), and the 10,725-parameter recurrent shared-proposal policy of D108/D109/D158. The pattern across all of them: capacity was never the binding constraint — representation and objective were. What survived: dense exact-value *teachers* over these encodings (D113, D152), the proposal-union action basis, and the rule that value must be computed at decision time (rollouts) rather than fitted into a snapshot scorer.

18 Why CPU and GPU training never agreed

The YT benchmark that decided where training runs (Arc A, D11 era) found the GPU path **9.8× faster** (9,769 vs 995 transitions/s) — and rejected it anyway: the same frozen hybrid-chopper evaluation scored **52/61 on CUDA versus 56/61 on CPU**, a 6.557-point gap against the preregistered 5-point parity cap, while every other parity metric passed. This is not a bug in either backend; it is the expected physics of the situation:

- **Floating-point addition is not associative.** GPU kernels (cuDNN convolutions, parallel reductions, tensor-core matmuls) sum in different orders than CPU BLAS, and may use different intermediate precisions (TF32, fused multiply-adds), so identical models on identical inputs produce logits differing in the last bits.

- **Reinforcement learning amplifies last-bit noise.** A last-bit logit difference flips an occasional near-tie action; a flipped action changes the trajectory; the changed trajectory changes the training data for every subsequent update. Unlike supervised learning — where the same noise stays bounded — the RL feedback loop compounds it into *macroscopically different policies*. That is exactly the observed signature: bitwise parity on static metrics, divergence only in rolled-out policy outcomes.
- **The same mechanism appears CPU-side with threads.** A 20-thread fit was not byte-stable for one seed (D117), and one threaded model hash differed at 2.4e-5 drift (D153) — parallel summation order again. Hence the house rules: **one deterministic training thread** for anything selected by byte-identity, thread-parallelism only for *evaluation* whose outputs are compared field-by-field, and byte-identical 1-vs-20 repeats as a standing integrity gate.

The project chose reproducibility over throughput: the sole D11 training run went to local CPU, and the frozen local/YT parity gate remains a precondition before any GPU result may be *selected* (YT stayed in use for exactly-replayable map/reduce simulation, where byte-parity was achieved and verified per shard).

19 Glossary — mother, crop, orchard

Terms that recur throughout the ledger, grounded in the game's mechanics and the resident's own source code.

- **Mother** — a tree the bot deliberately plants (or adopts) and then *keeps alive as a renewable seed source* instead of chopping it. Mechanically: a living tree bears species-typed fruit; a picked fruit can be replanted as a seed of that species; chopping the tree instead yields one-time wood. A mother is therefore reproductive capital — its recurring fruit income funds new plantings — where an ordinary tree is harvestable capital. The concept is first-class in the live resident: its scarce-map planner runs the intent chain `NeedSeed → HarvestSeed → PlantMother → TendMother → PlantCrop{mother, target}` (see `rust/src/bin/yamo_orchard_live.rs`).
- **Crop** — a tree planted for *conversion*: grown, then felled for wood/score. In the mother/crop loop, the mother's fruit becomes the seeds; the crops are her children. A **lineage** is the family of trees descending from one seed source; "lineage extinction" (a denial concept, class d) means no living tree of that family remains.
- **The mother/crop loop** — the self-renewing economy: protect one mature parent, harvest her fruit, plant children, fell the children at maturity, repeat. On tree-sparse maps this loop is the lifeline (the Gold-era deforestation stall was fixed by a seed reserve for exactly this reason). Its cost side: tending a mother consumes worker turns, and a mature *shared* mother's fruit can be captured by the opponent — the Phase 1–5 verdict that closed six aggressive mother/crop variants was precisely "action cost plus opponent capture exceed private crop value."
- **Orchard / secure orchard** — the resident's wrapper that runs this loop on favorable geometry: cells where the mother and her crops sit in resident-controlled territory (the promoted 07-17 stack's "secure-orchard coverage" widened exactly this geometry). "Releasing" the mother — treating her as a spare resource for other tasks — is closed: the ledger's verdict is that **the mother is a saturated producer, not an idle reservation** — her fruit throughput is already fully consumed by the loop, so liquidating or re-tasking her trades recurring apples for less wood than they are worth.
- **Where this touches the current thread** — the D164–D168 successor returns are the same economics seen from the worker's side: a producer leaves to suppress, and comes

back by planting a *new crop generation* from a banked seed (D167's BANK_SEED). The deposited bank those seeds come from is fed by harvests — including mother fruit — so the mother/crop loop is the upstream supply of the very returns D168 is now testing.

20 Where the records live

- Ledger vol 1 (Phases + D1–D166, frozen): `legend-top3-experiment-cycle-2026-07-18.md`.
- Ledger vol 2 (live): `legend-top3-experiment-cycle-vol2-2026-07-23.md`.
- Per-experiment records: `data/analysis/live-agent-6553250/dNNNa--{protocol,lock,result}`.
- Closed branches by topic: `docs/CONSTRAINTS.md` (classes a–h). Live state: `docs/STATE.md`.
- Priorities and gates: `docs/BACKLOG.md`.